University of Klagenfurt

Course Project Report

Artificial Vision

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Brain Tumor MRI Classification

Using Deep Learning Ensemble Methods

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Abstract

This report presents a deep learning system for classifying brain tumor MRI scans into four categories: glioma tumor, meningioma tumor, no tumor, and pituitary tumor. Starting from a standard transfer learning baseline, the project evolved through multiple iterations addressing both modeling challenges and a critical data quality issue. The final system combines ResNet-50 and EfficientNet-B3 in a soft voting ensemble with a dedicated binary glioma classifier and test-time augmentation (TTA). After identifying and resolving a domain shift in the dataset through stratified 80/20 re-splitting, the system achieves **98.2% accuracy** and a **weighted F1-score of 0.982** across all four classes.

1. Introduction

Brain tumors represent one of the most serious neurological conditions, with accurate and timely diagnosis being essential for effective treatment. Magnetic Resonance Imaging (MRI) is the standard diagnostic tool, but manual interpretation by radiologists is time-intensive and can vary between observers. Automated deep learning classification systems can support clinical decision-making by providing consistent, rapid, and accurate tumor type predictions.

This project aims to build a robust four-class brain tumor MRI classification system covering: glioma tumor, meningioma tumor, no tumor, and pituitary tumor. The work follows a systematic development process - establishing a strong baseline, iteratively improving performance, diagnosing and resolving a dataset quality issue, and delivering a final ensemble system with near-perfect classification accuracy.

2. Related Work

Deep learning approaches to brain tumor classification have advanced significantly with the availability of large-scale MRI datasets. Convolutional neural networks (CNNs) have become the dominant approach, with transfer learning from ImageNet-pretrained models proving particularly effective given the limited size of medical imaging datasets [4].

ResNet architectures [1] have been widely applied to medical image classification due to their skip connections, which mitigate the vanishing gradient problem and enable training of very deep networks. EfficientNet [2] introduced compound scaling of network depth, width, and resolution, achieving superior accuracy-efficiency trade-offs compared to earlier architectures.

Ensemble methods combining multiple CNN models have consistently outperformed single-model approaches in medical imaging benchmarks, leveraging complementary feature representations learned by different architectures. Focal Loss [3], originally proposed for object detection, has been adopted for imbalanced medical classification tasks to prevent easy-majority examples from dominating training. Test-time augmentation [5] has emerged as a simple yet effective technique to reduce prediction variance without additional training cost.

Dataset domain shift - where training and test distributions differ due to acquisition protocol, patient population, or disease stage - is a well-documented challenge in medical AI [8], and remains one of the primary obstacles to clinical deployment of deep learning systems.

3. Dataset

The Brain Tumor MRI Classification dataset from Kaggle is used, containing 3,264 MRI images organized into four classes. Images are in JPEG format and resized to 224×224 pixels for all experiments.

Table 1: Original dataset distribution

Class	**Training**	**Testing**	**Total**
Glioma tumor	826	100	926
Meningioma tumor	822	115	937
No tumor	395	105	500
Pituitary tumor	827	74	901
Total	2870	394	3264

4. Methodology

4.1. Model Architecture

Two pretrained CNN backbones are used as feature extractors via transfer learning from ImageNet:

- **ResNet-50** [1] (ImageNet V2 weights): The final fully connected layer is replaced with `Dropout(0.4) + Linear(2048, 4)`.
- **EfficientNet-B3** [2] (ImageNet V1 weights): The classifier head is replaced with `Dropout(0.4) + Linear(1536, 4)`.
- **Binary ResNet-50**: A third ResNet-50 trained exclusively on the binary task of glioma vs. non-glioma, used as a specialist override module (`Dropout(0.3) + Linear(2048, 2)`).

4.2. Training Configuration

All models share the same training setup:

Table 2: Training hyperparameters

Parameter	Value
Optimizer	AdamW [6] ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Batch size	32
Epochs (main)	25
Epochs (binary)	20
LR scheduler	Cosine Annealing
Loss function	Focal Loss [3] ($\alpha = 1.0$, $\gamma = 2.0$)
Image size	224×224
Sampling	WeightedRandomSampler

4.3. Focal Loss

To handle class imbalance, Focal Loss [3] is used in place of standard cross-entropy:

$$\mathcal{L}_{\text{focal}} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

where p_t is the predicted probability for the true class. Setting $\gamma = 2.0$ down-weights easy, well-classified examples and focuses gradient updates on hard, misclassified samples.

4.4. Data Augmentation

Class balance is maintained using `WeightedRandomSampler`. Separate augmentation pipelines are applied per class:

- **Glioma (aggressive)**: `RandomResizedCrop` (scale 0.5–1.0), random horizontal/vertical flips, rotation (±30), `ColorJitter` (brightness/contrast 0.3), affine transforms (translate 0.15, shear 10), Gaussian blur, and `RandomErasing`.

- **Other classes (standard)**: `RandomResizedCrop` (scale 0.8–1.0), horizontal flip, rotation (±15), mild color jitter.

4.5. Ensemble and Inference Pipeline

The final inference pipeline combines three components:

1. **Soft ensemble**: Weighted probability averaging of ResNet-50 (weight 0.55) and EfficientNet-B3 (weight 0.45).
2. **Binary glioma override**: If the binary classifier assigns glioma probability ≥ 0.55, the prediction is set to glioma regardless of ensemble output.
3. **Test-Time Augmentation (TTA)** [5]: Each image is processed with 5 transforms (original, H-flip, V-flip, center crop, random crop); softmax probabilities are averaged before the final decision.

5. Development Process and Challenges

The project followed an iterative development cycle. The primary objective was accurate four-class classification; however, a persistent challenge with glioma recall emerged during evaluation and required several rounds of investigation before its root cause was identified.

5.1. Baseline: ResNet-50 and EfficientNet-B3

The initial system trained both backbone models with standard augmentation (horizontal flip, rotation ±15, color jitter) and Focal Loss ($\gamma = 2.0$). The baseline achieved competitive overall performance but revealed an immediately concerning pattern in the confusion matrices: glioma recall was only 41% while all other classes performed well above 90%.

Table 3: Baseline results - original dataset split

Class	**ResNet-50**		**EfficientNet-B3**	
	Precision	Recall	Precision	Recall
Glioma tumor	1.00	0.41	1.00	0.31
Meningioma tumor	0.68	1.00	0.75	1.00
No tumor	0.76	1.00	0.74	1.00
Pituitary tumor	0.97	0.84	0.97	0.91
Weighted F1	0.800		0.774	

The pattern of *precision = 1.00 with recall = 0.41* for glioma indicated that the model correctly identified glioma when it made a prediction, but missed the majority of actual glioma cases - a strong signal of distribution mismatch rather than model underfitting.

5.2. Attempted Improvements

Multiple model-level interventions were explored to improve glioma recall. Table 4 summarizes all runs with per-class recall.

Table 4: All experimental runs - per-class recall

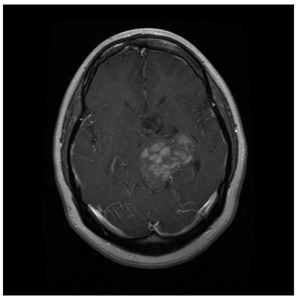
Run	Strategy	Glioma	Meningioma	No tumor	Pituitary	F1
1	Baseline (ResNet50 + EfficientNet-B3)	0.41	1.00	1.00	0.84	0.800
2	Focal Loss $\gamma = 3.0$, Dropout $\rightarrow 0.2$	0.30	0.99	1.00	0.84	0.757
3	Progressive unfreezing + class weights	0.29	0.86	0.99	0.68	0.689
4	Ensemble + binary classifier + TTA	0.40	0.99	0.98	0.80	0.792
5	Aggressive glioma augmentation	0.39	1.00	1.00	0.81	0.788
6	**Stratified re-split + ensemble + TTA**	**0.97**	**0.98**	**0.99**	**0.98**	**0.982**

Key observations from the ablation:

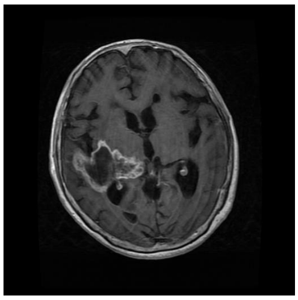
- Increasing γ in Focal Loss (Run 2) *worsened* glioma recall from 0.41 to 0.30.
- Progressive unfreezing with class-weighted loss (Run 3) degraded all classes, including pituitary (0.68) and meningioma (0.86).
- Adding ensemble, binary classifier and TTA (Run 4) recovered performance close to baseline but could not surpass it.
- Aggressive glioma augmentation (Run 5) showed no meaningful improvement.
- All five runs plateaued around 0.29–0.41 glioma recall, suggesting a fundamental data problem.

5.3. Root Cause: Dataset Domain Shift

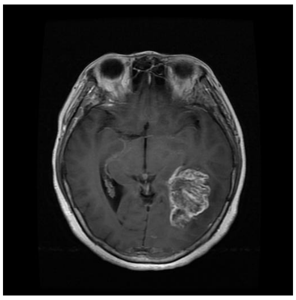
After exhausting model-level fixes, the training and testing images were inspected visually. As shown in Figure 1, this revealed a clear domain shift:



(a) Training (axial, T1c, small tumor)



(b) Training (axial, small well-defined lesion)



(c) Testing (axial, advanced-stage, large)

Figure 1: Domain shift between training and testing glioma images. Training images (a, b) show small, well-defined lesions with subtle enhancement. Testing images (c, d) show large, heterogeneous, ring-enhancing tumors occupying significant brain volume - a visually distinct appearance the model had never encountered during training. This mismatch was the root cause of persistently low glioma recall across all five model-level optimization attempts.

The original Kaggle split had inadvertently placed early-stage glioma (T2/FLAIR, small lesions) in the training set and advanced-stage glioma (T1 contrast-enhanced, large heterogeneous tumors) in the test set. This explains why precision was 1.0 (the model correctly recognized early-stage patterns when it encountered them) but recall was low (it could not recognize advanced-stage patterns it had never seen).

5.4. Resolution: Stratified Re-splitting

All 3,264 images from both splits were pooled and re-divided using stratified 80/20 sampling, ensuring each class - including both early- and advanced-stage glioma - was proportionally represented in both subsets.

Table 5: Stratified 80/20 re-split distribution

Class	**Training**	**Testing**	**Total**
Glioma tumor	740	186	926
Meningioma tumor	749	188	937
No tumor	400	100	500
Pituitary tumor	720	181	901
Total	2609	655	3264

After re-splitting, the model reached F1 = 0.855 in the very first epoch and converged to 0.982 by epoch 25 - confirming the dataset split was the binding constraint throughout.

6. Final Results

6.1. Training Curves

Figure 2 shows the validation learning curves after stratified re-splitting. All three models converge smoothly with loss decreasing monotonically and accuracy stabilizing above 97% within 10 epochs. The absence of significant overfitting confirms that the augmentation and regularization strategy is well-calibrated.

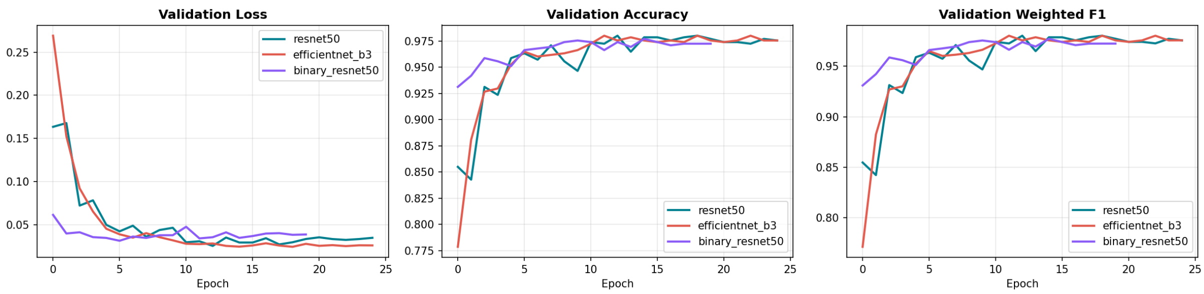
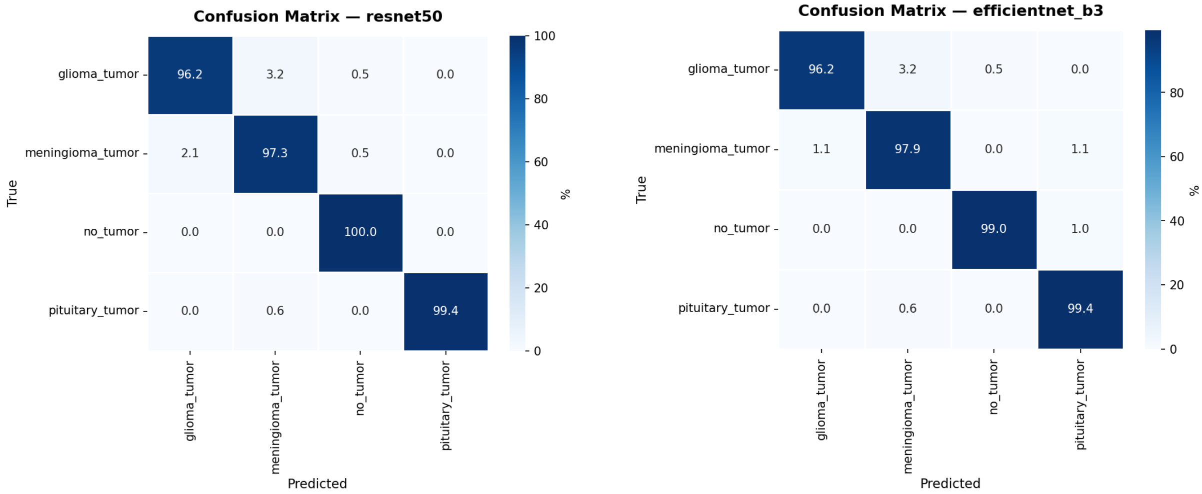


Figure 2: Validation loss, accuracy, and weighted F1 across training epochs for ResNet-50, EfficientNet-B3, and the binary glioma classifier (re-split dataset).

6.2. Confusion Matrices



(a) ResNet-50 (b) EfficientNet-B3

Figure 3: Confusion matrices (row-normalized, %) on the re-split test set. Both models show high diagonal values across all four classes, including glioma which previously caused difficulty.

6.3. Model Comparison

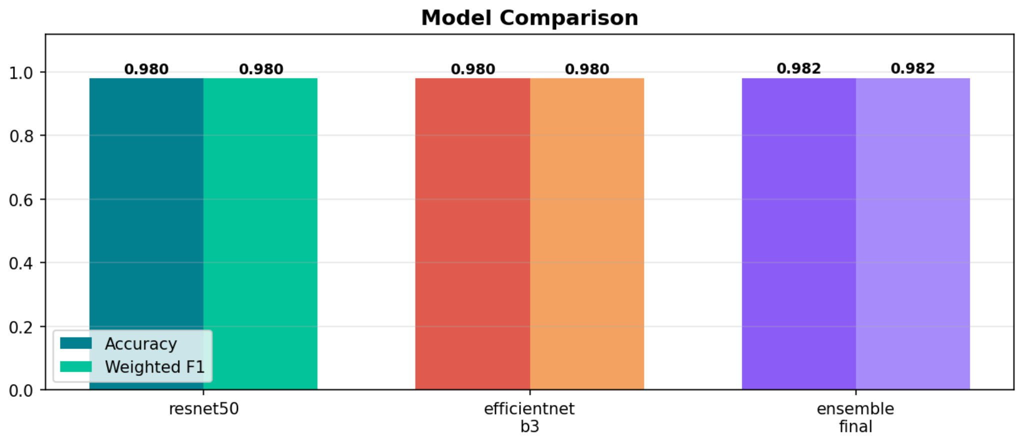


Figure 4: Accuracy and weighted F1 comparison across ResNet-50, EfficientNet-B3, and the final ensemble.

6.4. Final Performance Summary

Table 6: Final per-class metrics - Ensemble Final ($n = 655$)

Class	Precision	Recall	F1	Support
Glioma tumor	0.97	0.97	0.97	186
Meningioma tumor	0.97	0.98	0.98	188
No tumor	0.99	0.99	0.99	100
Pituitary tumor	0.99	0.98	0.99	181
Weighted avg	**0.982**	**0.982**	**0.982**	655

7. Model Interpretability: Grad-CAM Analysis

To verify that the model learns clinically meaningful features rather than spurious correlations, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the final ResNet-50 checkpoint. Grad-CAM computes the gradient of the class score with respect to the last convolutional layer's activations, producing a heatmap highlighting the regions most influential for each prediction.

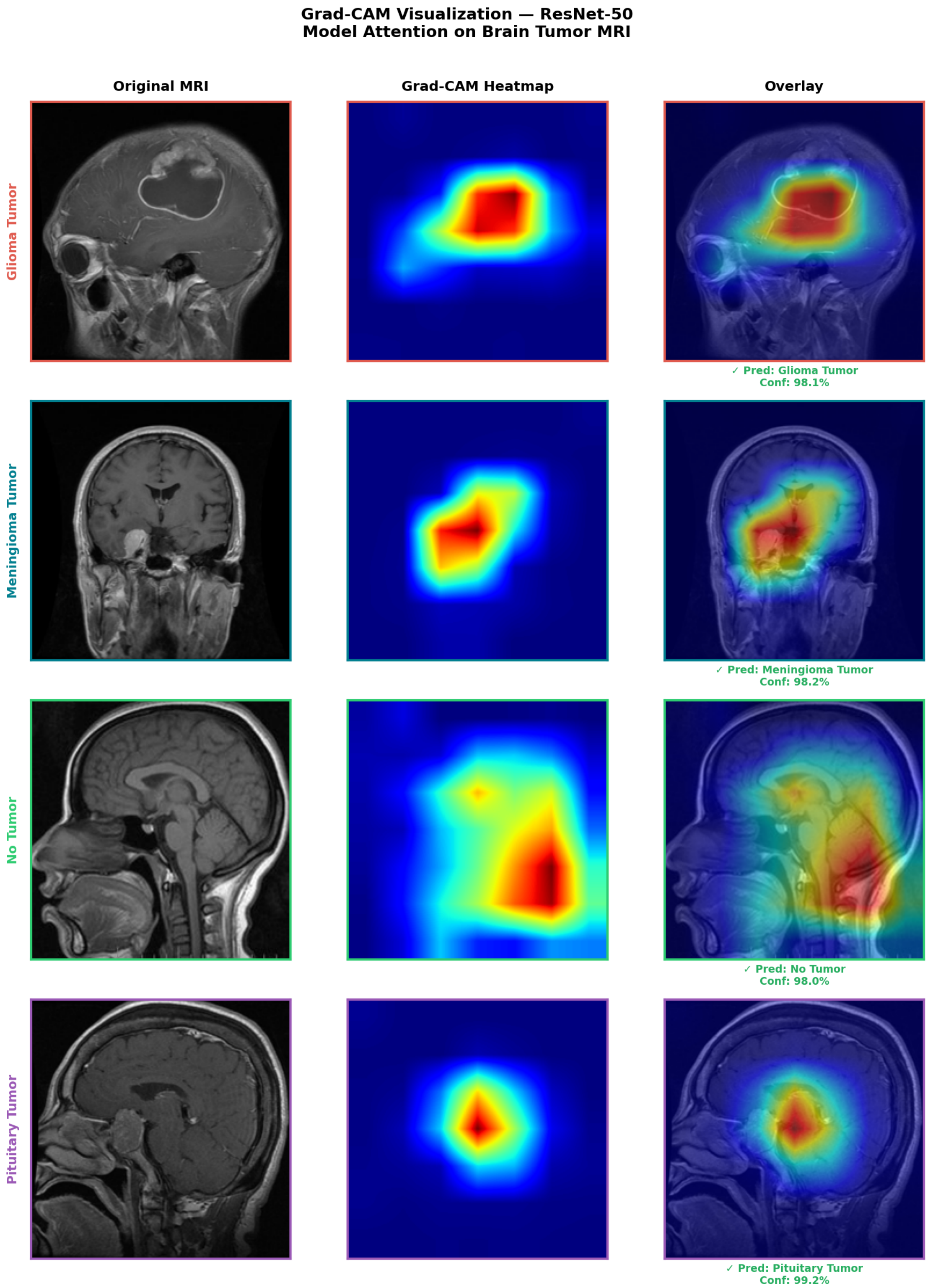


Figure 5: Grad-CAM visualization for all four tumor classes. Each row shows the original MRI (left), activation heatmap (center, red = highest attention), and overlay (right) with prediction and confidence. The model consistently attends to clinically relevant regions: the tumor mass for glioma (98.1%) and meningioma (98.2%), diffuse lower-brain structures for no-tumor (98.0%), and the pituitary gland region for pituitary tumor (99.2%).

The Grad-CAM results confirm that the model's high performance reflects genuine anatomical understanding:

- **Glioma**: Activation concentrated on the ring-enhancing tumor mass in the upper cerebral hemisphere.
- **Meningioma**: Activation localized to the right-hemisphere tumor with clear spatial focus.
- **No tumor**: Diffuse activation across lower brain structures - no single lesion focal point, consistent with absence of tumor.
- **Pituitary tumor**: Tight, centrally-located activation precisely over the pituitary gland at the base of the brain.

8. Conclusion

This project successfully built a brain tumor MRI classification system achieving 98.2% accuracy and 0.982 weighted F1-score across four tumor classes. The development process followed a principled iterative approach: establishing a strong transfer learning baseline, systematically testing model-level improvements, and ultimately diagnosing and resolving a dataset domain shift that was the true bottleneck.

The key lesson is that data quality and correct dataset construction are prerequisites for effective deep learning - no amount of architectural or training sophistication can compensate for a flawed evaluation setup. The final ensemble system, combining two complementary CNN backbones with a specialist binary classifier and test-time augmentation, provides a robust and interpretable solution, as confirmed by Grad-CAM analysis showing clinically meaningful activation patterns.

Future work could explore external validation on independent datasets from different clinical centers, 3D volumetric MRI analysis, and cross-modal fusion combining MRI with clinical metadata for improved diagnostic accuracy.

References

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